📘 AI/ML Internship – Day 80 Notes & Tasks

🟢 Module 9: Deploying a RAG Chatbot – From Development to Production

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📌 Part 1: Recap of Day 79

Yesterday we learned:

- ✅ RAG Evaluation
- ✅ Precision
- ✅ Recall
- ✅ Hallucination
- ✅ Faithfulness
- ✅ Evaluation Frameworks

Today we will learn:

- What is Deployment?
- Why Deployment is Important
- RAG Deployment Architecture
- FastAPI Integration
- Frontend Integration
- Deployment Platforms
- Monitoring & Maintenance

📌 Part 2: What is Deployment?

Definition

Deployment is the process of making an AI application available for real users after development and testing.

Simple Definition

Building a chatbot on your computer is called development.

Making it accessible to users through a website or application is called deployment.

Example

Before Deployment:

- Chatbot runs only on your laptop.
- Only the developer can use it.

After Deployment:

- Users can access it through:
    - Website
    - Mobile App
    - Company Portal
    - API

📌 Part 3: Why Do We Need Deployment?

Imagine a company builds an HR chatbot.

If it stays on the developer's computer:

❌ Employees cannot use it.

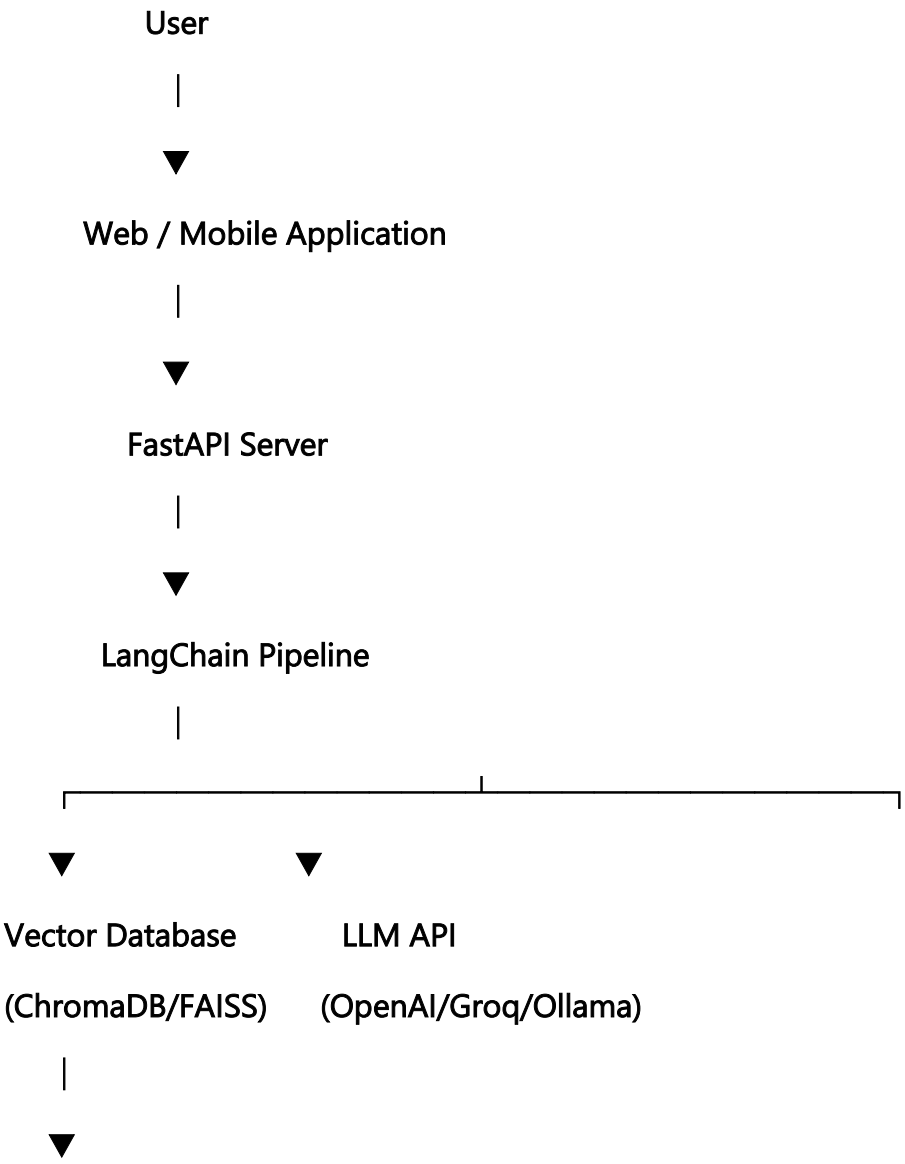
After deployment:

✅ Every employee can access it.

Benefits:

- 24×7 availability
- Remote access
- Centralized updates
- Scalable for many users

📌 Part 4: RAG Deployment Architecture



📌 Part 5: Components of a Production RAG System

1. Frontend

Provides the user interface.

Examples:

- HTML/CSS
- Bootstrap
- JavaScript

- React
- Flutter

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2. Backend

Handles requests and communicates with the AI.

Examples:

- FastAPI
- Flask
- Django

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3. Vector Database

Stores embeddings.

Examples:

- ChromaDB
- FAISS
- Pinecone
- Weaviate

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4. LLM

Generates answers.

Examples:

- OpenAI GPT
- Groq
- Ollama
- Gemini

📌 Part 6: FastAPI Integration

Typical API endpoints:

Endpoint	Purpose
/upload	Upload PDF documents
/ask	Ask questions to the chatbot
/history	Retrieve chat history
/reset	Clear conversation memory

Example Request

POST /ask

```
{
  "question":"What is Machine Learning?"
}
```

Example Response

```
{
  "answer":"Machine Learning is a branch of AI...",
  "source":"ML_Notes.pdf",
  "page":15
}
```

📌 Part 7: Deployment Platforms

Popular deployment platforms:

Platform	Purpose
Render	Simple cloud deployment
Railway	Backend hosting
Hugging Face Spaces	AI application hosting
AWS	Enterprise cloud platform
Azure	Microsoft cloud services
Google Cloud Platform (GCP)	Cloud infrastructure
Docker	Containerized deployment
Kubernetes	Large-scale deployment

📌 Part 8: Monitoring After Deployment

Deployment is not the end.

The system should continuously monitor:

- Number of users
- Response time
- Errors
- Failed requests
- API usage
- User feedback

📌 Part 9: Logging

Logs help developers identify issues.

Example Log:

Time: 10:30 AM

User: user123

Question: What is AI?

Response Time: 1.2 seconds

Status: Success

Logs are useful for debugging and performance analysis.

📌 Part 10: Security Best Practices

Always:

- Use HTTPS
- Protect API keys
- Authenticate users
- Validate uploaded files
- Limit upload size
- Prevent prompt injection attacks
- Restrict unauthorized access

📌 Part 11: Scaling a RAG System

As more users access the chatbot:

Challenges:

- Increased traffic
- Higher API costs
- More storage
- Slower response time

Solutions:

- Load balancing
- Caching
- Optimized retrieval

- Distributed databases
- Horizontal scaling

📌 Part 12: Real-World Applications

Customer Support Portal

Employees ask company-related questions.

Educational Portal

Students upload notes and ask questions.

Hospital Assistant

Doctors search medical guidelines.

Legal Assistant

Lawyers search contracts and legal documents.

Enterprise Knowledge Base

Employees search company documentation.

📌 Part 13: Advantages of Deployment

✅ Accessible from anywhere.

✅ Supports multiple users.

✅ Easy updates.

✅ Centralized management.

✅ Production-ready.

📌 Part 14: Limitations

❌ Cloud hosting costs.

❌ API costs.

❌ Requires monitoring.

❌ Security must be maintained.

❌ Infrastructure management is required.

📌 Part 15: Theory Questions

1. What is deployment?

Deployment is the process of making an application available for end users after development and testing.

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2. Why is deployment important?

It allows users to access the AI application through websites, mobile apps, or APIs.

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3. Name four deployment platforms.

- Render
- Railway
- AWS
- Hugging Face Spaces

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4. Why is logging important?

Logging records application events, helping developers identify bugs, monitor performance, and troubleshoot issues.

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5. Why should API keys be protected?

API keys provide access to AI services. If exposed, they can lead to unauthorized usage and additional costs.

📌 Part 16: Interview Questions

Explain the architecture of a deployed RAG chatbot.

A deployed RAG chatbot consists of a frontend for user interaction, a backend (such as FastAPI) for processing requests, a vector database for retrieving relevant document embeddings, and an LLM for generating responses. The backend coordinates communication between these components.

What is the role of FastAPI in RAG deployment?

FastAPI receives user requests, processes uploaded documents, retrieves relevant information from the vector database, sends context to the LLM, and returns the generated response.

Why is monitoring important after deployment?

Monitoring helps detect failures, measure performance, track user activity, and ensure the chatbot remains reliable.

How can a deployed RAG chatbot be secured?

By using HTTPS, authentication, secure API key storage, input validation, file upload restrictions, and protection against prompt injection.

Name three cloud platforms used for AI deployment.

- AWS
- Microsoft Azure
- Google Cloud Platform (GCP)

🎯 Tasks for Day 80

🧠 Theory

1. Define deployment.
2. Explain the architecture of a production RAG chatbot.
3. What is the role of FastAPI?
4. Why is monitoring necessary?
5. List five security best practices for AI deployment.

💻 Practical Task 1

Create a FastAPI endpoint that accepts a user question and returns an AI-generated answer.

💻 Practical Task 2

Create an endpoint to upload PDF documents and store their embeddings in ChromaDB.

💻 Practical Task 3

Design a simple HTML page where users can:

- Upload PDF files.
- Ask questions.
- View chatbot responses.
- View source document names.

💻 Practical Task 4

Add logging to your chatbot application to record:

- User questions
- Response times
- Success or failure status

💻 Practical Task 5

Research one cloud deployment platform (Render, Railway, AWS, Azure, or GCP) and prepare a short report covering:

- Features
- Advantages
- Limitations
- Pricing model